ai.py - C:\Users\ganes\AppData\Local\Programs\Python\Python313\ai.py (3.13.3)

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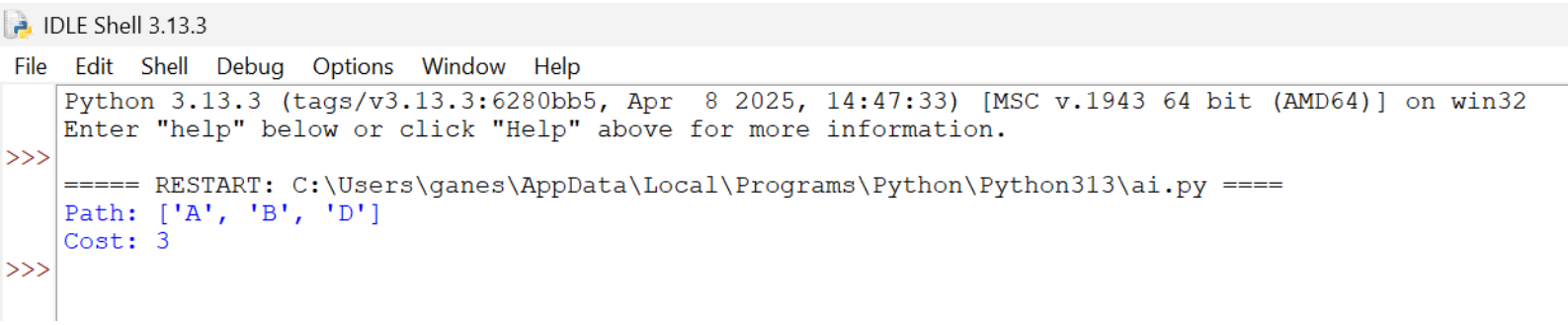
```
import heapq

def astar(graph, heuristic, start, goal):
    pq = [(heuristic[start], 0, start, [start])]
    visited = set()

    while pq:
        f, g, node, path = heapq.heappop(pq)

        if node == goal:
            return path, g

        if node in visited:
            continue
        visited.add(node)

        for neighbor, cost in graph[node]:
            if neighbor not in visited:
                new_g = g + cost
                heapq.heappush(
                    pq, (new_g + heuristic[neighbor],
                         new_g, neighbor, path + [neighbor])
                )
    return None, float('inf')

graph = {
    'A': [('B', 1), ('C', 4)],
    'B': [('D', 2)],
    'C': [('D', 1)],
    'D': []
}
heuristic = {'A': 3, 'B': 2, 'C': 1, 'D': 0}

path, cost = astar(graph, heuristic, 'A', 'D')
print("Path:", path)
print("Cost:", cost)
```


IDLE Shell 3.13.3
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Python 3.13.3 (tags/v3.13.3:6280bb5, Apr 8 2025, 14:47:33) [MSC v.1943 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:\Users\ganes\AppData\Local\Programs\Python\Python313\ai.py ====
Path: ['A', 'B', 'D']
Cost: 3
>>>